



UNIVERSITÄT
LEIPZIG

Medizinische Fakultät

Architecture of IS in GW / Information management in clinical research

Transformation of a CSV data set to HL7 FHIR

Leipzig, date

imise.

CONTENTS

1. Background

- a. Communication server
- b. FHIR

2. NextGen Connect (formerly Mirth Connect)

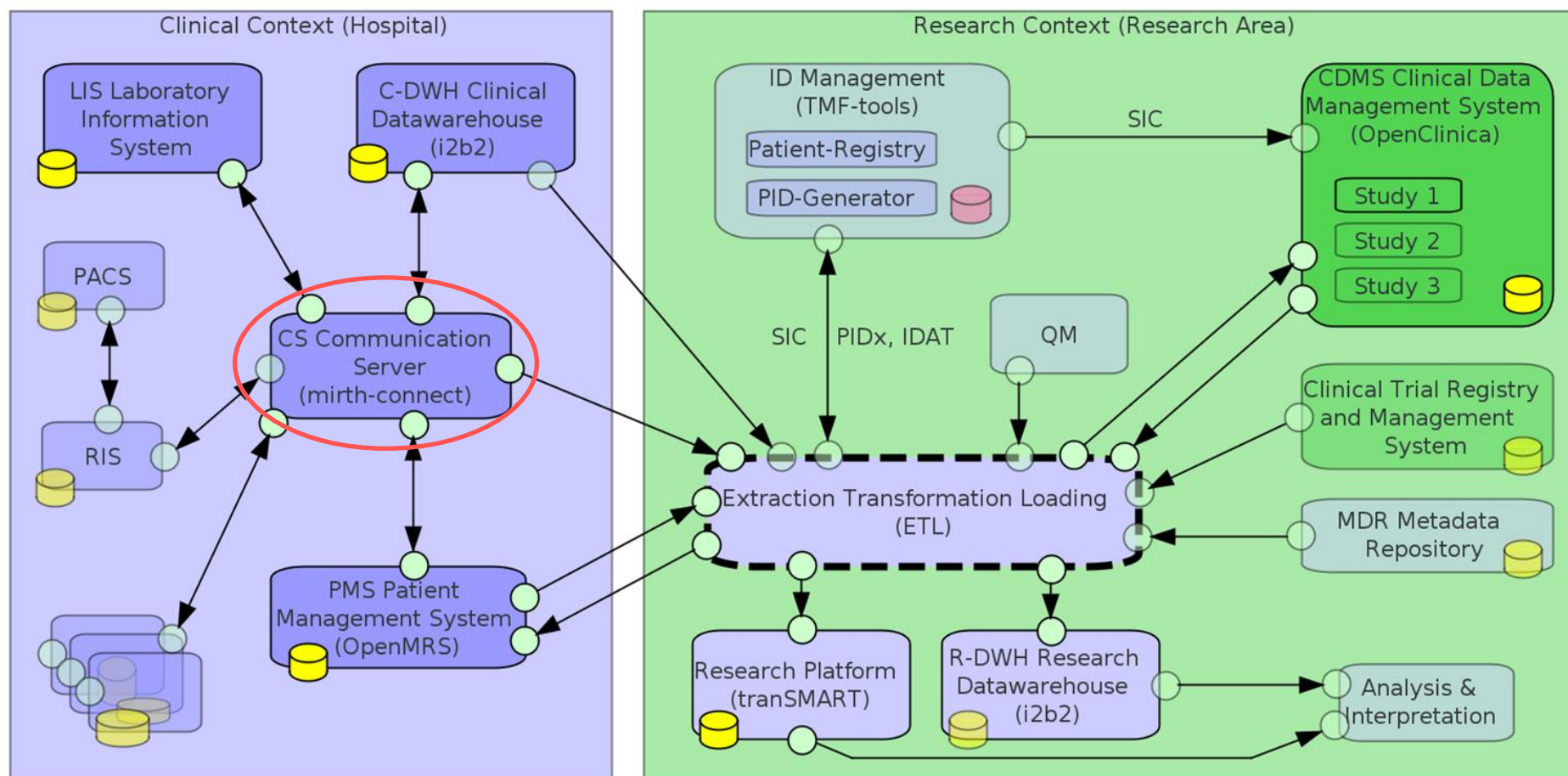
- a. Transformation process
- b. FHIR templates
- c. Channels
- d. Filters & Transformers

3. Exercise

COMMUNICATION SERVER

- **Function**
 - Central distribution of messages
 - Filter
 - Transformation
 - Avoidance of $n*m$ communication interfaces
- **Software products:**
 - eGate
 - Kodix (UKL)
 - **NextGen Connect** (formerly Mirth Connect)

COMMUNICATION SERVER



FAST HEALTHCARE INTEROPERABILITY RESOURCES (FHIR)

- Consisting of consistent, easy-to-understand resources
- 80-20 principle: the standard (currently R4) covers most applications, with the rest covered by extensions
- Web technology (Restful API) + human readability
- Open source (specification, tools, servers, libraries)
- Maintained by HL7
- Used in Germany as part of the MII (core data set)

Patient	I	Patient
id	 Σ	0..1 string
meta	 Σ	0..1 Meta
identifier	 Σ	0..* Identifier
active	Σ ?!	0..1 boolean
name	 Σ	0..* HumanName
telecom	Σ I	0..* ContactPoint
gender	 Σ	0..1 code Binding
birthDate	 Σ	0..1 date
deceased[x]	 Σ ?!	0..1
address	 Σ	0..* Address
maritalStatus		0..1 CodeableConcept Binding
multipleBirth[x]		0..1
photo	I	0..* Attachment
contact	I	0..* BackboneElement
communication		0..* BackboneElement
generalPractitioner	I	0..* Reference(Organization Practitioner Practit...
managingOrganization	Σ I	0..1 Reference(Organization)
link	 Σ ?!	0..* BackboneElement

https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Person_Version_2/MIIGModulPerson-TechnischImplementierung-FHIR-Profile-PatientInPatient.html

PURPOSE OF THE EXERCISE

- Transformation from CSV to FHIR
 - Mapping to FHIR with MII-KDS
- Creating channels with Mirth-Connect
 - Receiving data
 - Transforming
 - Outputting data in FHIR format
- Sending FHIR data to a FHIR server
 - Verifying the transformed FHIR data with ClinFHIR

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

PROCESS

1. Input data
 - CSV data set
2. Transformation with Mirth-Connect
 - Mapping of CSV columns to FHIR attributes
 - Based on FHIR message template
3. Output
 - JSON bundle in FHIR format
 - For sending to a FHIR server

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

FHIR TEMPLATE: MII PATIENT

Patient	I	Patient
id	S Σ 0..1	string
meta	S Σ 0..1	Meta
identifier	S Σ 0..*	Identifier
active	Σ ?! 0..1	boolean
name	S Σ 0..*	HumanName
telecom	Σ I 0..*	ContactPoint
gender	S Σ 0..1	code Binding
birthDate	S Σ 0..1	date
deceased[x]	S Σ ?! 0..1	
address	S Σ 0..*	Address
maritalStatus	0..1	CodeableConcept Binding
multipleBirth[x]	0..1	
photo	I 0..*	Attachment
contact	I 0..*	BackboneElement
communication	0..*	BackboneElement
generalPractitioner	I 0..*	Reference(Organization Practitioner Practit...
managingOrganization	Σ I 0..1	Reference(Organization)
link	S Σ ?! 0..*	BackboneElement



Template for FHIR messages to be sent

Outbound Message Template

Data Type:

```
{
  "resourceType": "Bundle",
  "type": "transaction",
  "entry": [
    {
      "resource": {
        "resourceType": "Patient",
        "meta": {
          "profile": [
            "https://www.medizininformatik-initiative.de/fhir/core/modul-person/"
          ]
        },
        "identifier": [
          {
            "use": "usual",
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                  "code": "MR"
                }
              ]
            }
          }
        ],
        "system": "http://mimic.mit.edu/fhir/mimic/identifier/patient",
        "value": "10038992"
      }
    },
    {
      "name": [
        {
          "use": "official",
          "family": "Patient_10038992"
        }
      ]
    }
  ]
}
```

Mirth-Connect Patient Source Transformer

https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Person_Version_2/MIIGModulPerson-

TechnischeImplementierung-FHIR-Profil-PatientInPatient.html

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

FHIR TEMPLATE: MII ENCOUNTER

Encounter	I	Encounter
id	S Σ	0..1 string
meta	S Σ	0..1 Meta
extension	I	0..* Extension
identifier	S Σ	0..* Identifier
status	S Σ ?!	1..1 code Binding
statusHistory		0..* BackboneElement
class	S Σ	1..1 Coding Binding
classHistory		0..* BackboneElement
type	S Σ	0..* CodeableConcept
serviceType	S Σ	0..1 CodeableConcept
priority		0..1 CodeableConcept
subject	S Σ I	1..1 Reference(Patient Group)
episodeOfCare	Σ I	0..* Reference(EpisodeOfCare)
basedOn	I	0..* Reference(ServiceRequest)
participant	Σ	0..* BackboneElement
appointment	Σ I	0..* Reference(Appointment)
period	S I	1..1 Period
length	I	0..1 Duration
reasonCode	Σ	0..* CodeableConcept Binding
reasonReference	Σ I	0..* Reference(Condition Procedure Observation...)
diagnosis	S Σ	0..* BackboneElement
account	I	0..* Reference(Account)
hospitalization	S	0..1 BackboneElement
location	S	0..* BackboneElement
serviceProvider	S I	0..1 Reference(Organization)
partOf	S I	0..1 Reference(Encounter)



Outbound Message Template

Data Type: JSON Properties

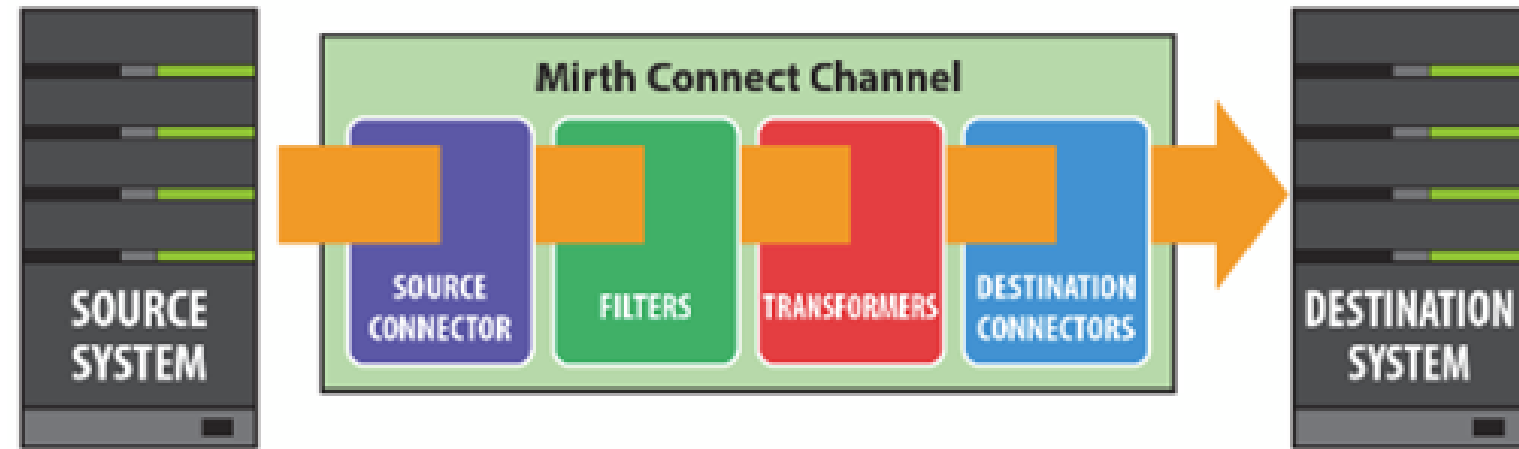
```
{
  "resourceType": "Bundle",
  "type": "transaction",
  "entry": [
    {
      "resource": {
        "resourceType": "Encounter",
        "meta": {
          "profile": [
            "https://www.medizininformatik-initiative.de/fhir/core/modul-fall/Stru"
          ]
        },
        "status": "finished",
        "identifier": [
          {
            "type": {
              "coding": [
                {
                  "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                  "code": "VN"
                }
              ]
            },
            "system": "http://mimic.mit.edu/fhir/mimic/identifier/encounter-hosp",
            "value": "25135483"
          }
        ],
        "class": {
          "code": "AMB",
          "display": "ambulatory",
          "system": "http://hl7.org/fhir/v3/ActCode/cs.html"
        },
        "subject": {
          "reference": "Patient/10004235"
        }
      }
    }
  ]
}
```

Code system: https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Fall_Version_2/MIIGModulFall-TechnischeImplementierung-FHIRProfile-EncounterKontaktGesundheitseinrichtung.html

https://www.medizininformatik-initiative.de/Kerndatensatz/Modul_Fall_Version_2/MIIGModulFall-TechnischeImplementierung-FHIRProfile-EncounterKontaktGesundheitseinrichtung.html

MIRTH-CONNECT

CHANNEL STRUCTURE



https://www.adldata.org/download/Manuals/Mirth_Connect_3.4_Users_Guide.pdf

1. **Source** (input)

- Read CSV data set
- File Reader (defined folder)

2. **Filters & Transformer** (processing)

- Transforming input data CSV to FHIR template
- Mapping CSV columns to FHIR attributes

3. **Destination** (output)

- Send the transformed FHIR JSON messages to the FHIR server (REST API)
- Http sender

MIRTH-CONNECT

2. SOURCE TRANSFORMER

- Transformation of the data set to FHIR
- Mapping CSV columns to FHIR attributes
- Transformer Step: attribute to be mapped

- **Input:** CSV file
- **Output:** FHIR message template

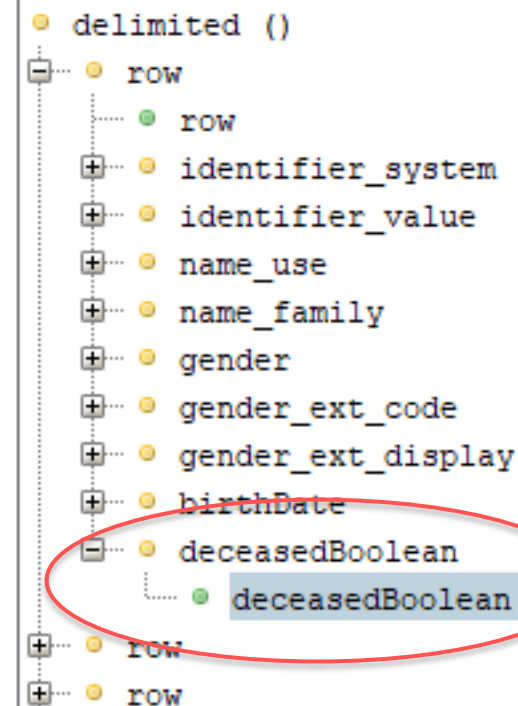
TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE

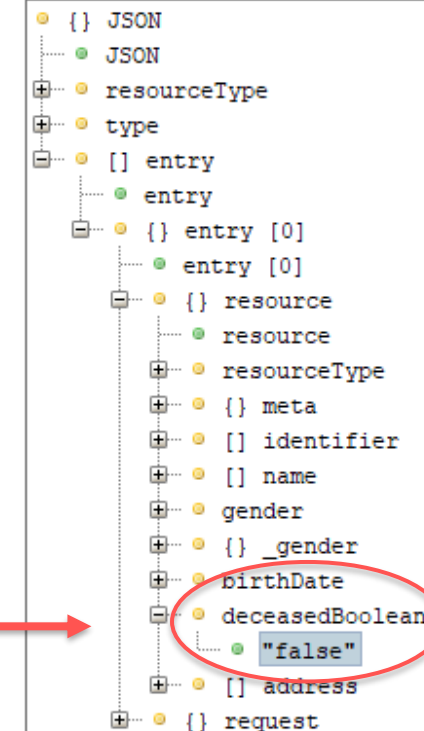
Input: CSV file

- All rows and columns as tree structure
- Mapping of the first row sufficient

Inbound Message Template Tree

Filter: 

Outbound Message Template Tree

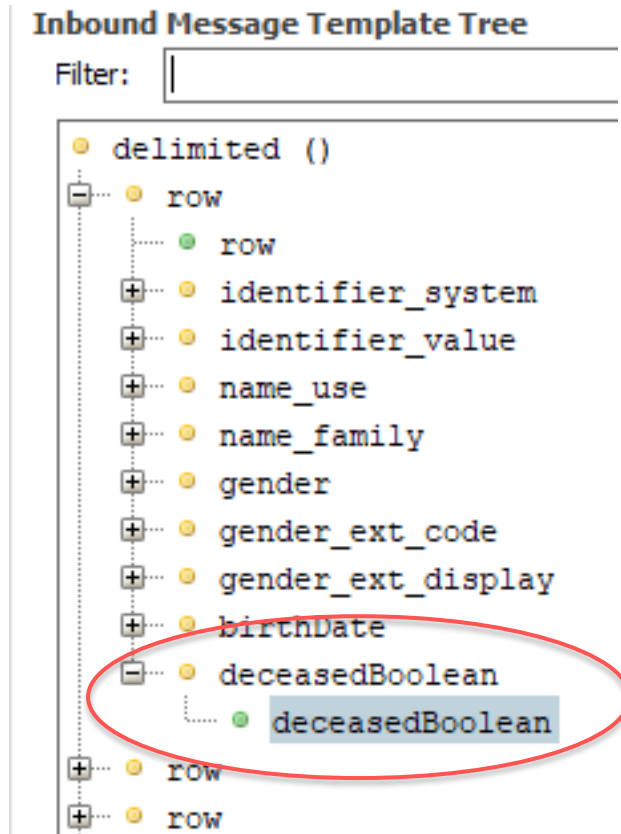
Filter: 

Output: FHIR JSON template

- A JSON tree

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE



- Mirth-Connect Transformer
- **Input CSV:**

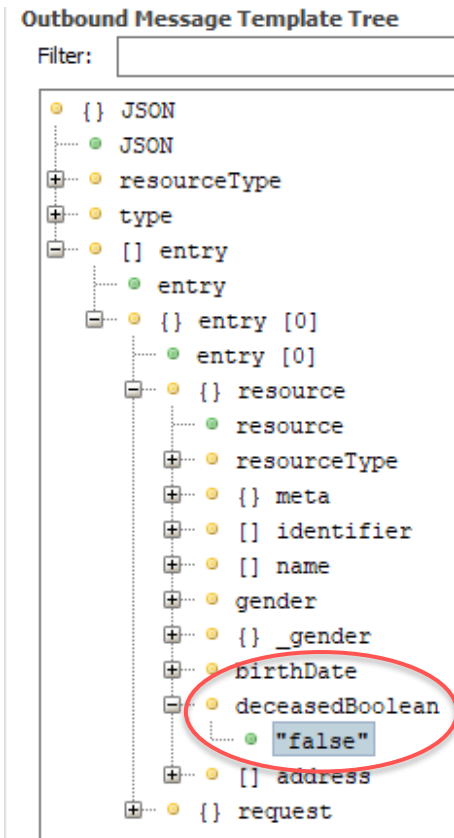
Mirth-Connect syntax

➤ `msg['row'][0]['deceasedBoolean'].toString()`

Target variable

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE



- Mirth-Connect Transformer
- **Output** FHIR message template:

Mirth-Connect Syntax

➤ `tmp['entry'][0]['resource']['deceasedBoolean']`

Target variable

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

TRANSFORMATION TERMS

- Transformer Step
 - 1 step for each attribute to be mapped
- Step Type: **Message Builder**
 - Replaces attribute in FHIR template with attribute from CSV data record
- **Message Segment** (Output)
 - Text field for attribute from **FHIR template**
- **Mapping** (Input)
 - Text field for attribute from **CSV data record**

TRANSFORMATION FROM CSV TO FHIR WITH MIRTH-CONNECT

MII PATIENT TRANSFORMATION: EXAMPLE

Enabled	#	Name	Type
☒	0	• identifier_system	Message Builder
☒	1	• identifier_value	Message Builder
☒	2	• name_use	Message Builder
☒	3	• gender_code	Message Builder
☒	4	• gender_display	Message Builder
☒	5	• birthdate	Message Builder
☒	6	• deceased	Message Builder

Step	Generated Script
Message Segment:	<code>tmp[entry][0][resource][identifier][0][system]</code>
Mapping:	<code>msg[row][0][identifier_system].toString()</code>

TASKS (SEE EXERCISE SHEET)

1. Mirth Connect

1.1 - 1.2

Create channel

1.3.1

Source connector

1.3.2

Destination connector

1.4 - 1.5

Extend transformer

1.6

Deploy channels and send FHIR messages

1. ClinFHIR

2.1

View transformed patients and cases

2.2 - 2.3

Answer questions

TASKS (SEE EXERCISE SHEET)

Exercise sheet in **GitHub**:

<https://github.com/IMISE/MI-Lab-E02-CSV-to-FHIR>

→ Under /Exercise/Exercise_sheet.pdf





UNIVERSITÄT
LEIPZIG

Medizinische Fakultät

THANK YOU!

Mona Perbix

Institute for Medical Informatics, Statistics, and Epidemiology (IMISE)